Build Your Own AI Research Assistant

Automate web scraping and summarization straight to your phone using n8n and OpenAI.

The Problem

Context switching kills your productivity flow state.

- Opening new browser tabs for every query.
- Sifting through Google search results and ads.
- Reading multiple long-form articles.
- Manually summarizing data to take action.

It takes too long to get a simple, factual answer.

The Solution

An autonomous agent living right inside your messaging app.

- **Instant Access:** Send a quick text message with your research query.
- **Live Search:** The AI browses the live internet, not just old training data.
- **Synthesis:** It reads the necessary data and texts you the exact, concise answer.
- **Zero Friction:** No tabs, no context switching, instant results directly to your phone.

The 3-Step Agent Loop

1. Trigger

The workflow sits idle, listening for a specific command or message sent from you in a designated Telegram chat.

2. Think & Act

An LLM Agent analyzes the request, decides what information is missing, and uses SerpAPI to fetch live data from the web.

3. Respond

The agent formats its findings into a clean, concise summary and routes it back to the exact chat ID it originated from.

The Tech Stack

Orchestration & UI

- n8n: The visual node-based builder that acts as the nervous system connecting everything.
- Telegram: The front-end mobile interface where you interact seamlessly with the bot.

Brains & Eyes

- OpenAI: The underlying reasoning engine (gpt-4o-mini) that understands complex queries.
- SerpAPI: The tool that grants the AI agent the ability to scrape Google search results in real-time.

Workflow Architecture

1. Webhook

Telegram Trigger Node receives the incoming text payload.

2. AI Agent

LangChain Agent processes the intent and plans the execution.

3. Tool Call

Agent triggers SerpAPI to fetch live Google search results.

4. Output

Telegram Send Node routes the final response back to the user.

Why This Stack?

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Total Nodes

Zero Code Required.

You can build this entire functional AI agent using just 4 visual nodes in n8n. It requires no Python, no deploying servers, and takes less than 15 minutes to configure from start to finish.

Let's Build It!

Switching to the n8n workspace...

Image Sources

https://media.easy-peasy.ai/40955941-c44d-4ec3-98a6-c752af81c537/07d7c413-af87-4d00-ab43-b2a596acdc8e.png

Source: easy-peasy.ai

https://media.easy-peasy.ai/217333b6-f002-407b-bab2-2f2786736b04/9486ff6d-81cb-4a13-86f3-f20b1d9aae68.png

Source: easy-peasy.ai